BITS-SIMS

STUDENT INFORMATION MANAGEMENT SYSTEM

Comprehensive System Fields, Module Specifications and Database Design Reference

Prepared from Scenario 9: Student Information System

BITS Officers 2026

Planning and analysis reference | 30 October 2026

Proposed fields and data structures are recommendations for system design.
Final policies, grading rules, retention periods, and access permissions should be confirmed with the institution.

Document Overview

This reference translates the BITS-SIMS problem scenario, requirements analysis, and supplied UI mockups into a detailed proposed field catalogue and implementation-oriented system design. It is intended to support database design, interface planning, role and permission definition, and development task breakdown.

Purpose and scope

The proposed system centralizes student information that is currently maintained through paper files and multiple departmental spreadsheets. It covers student identity and profiles, admission and enrollment, courses and class assignments, attendance, grades, official academic records, documents, reports, and system administration.

How to use this reference

- Use the field dictionaries as a starting point for database tables and form controls. They describe suggested fields, generic SQL data types, and the intended purpose of each field.
- Treat the four primary modules as the main navigation areas. Smaller functions such as Attendance Tracking, Grade Management, and Document Management are submodules, not necessarily separate top-level menu items.
- Use the access matrix as an initial role-permission proposal. Confirm it with school policy before implementation, especially for official grades, student records, and confidential documents.
- Use the validation rules and workflows to define backend checks. Hiding a button in the interface is not sufficient authorization; the server must enforce permissions too.
- Do not create a separate screen for every table. Many tables are supporting records maintained by the application in the background.

Document guide

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> **Important distinction** The data types are generic SQL recommendations, not a binding technology choice. Adapt syntax to the selected database engine. Use consistent primary-key and foreign-key types throughout the implementation.

1. System Architecture and Design Conventions

BITS-SIMS should be designed as a centralized student information system with a shared student identity and connected academic records. The main modules should read from and update the same underlying records according to the logged-in user's permissions.

1.1 Primary modules

Primary module	Main functions	Typical users
1. User Authentication and Access Control	Login, password recovery, account management, role assignment, permissions, session and security controls.	ICT Services, Education Technology Services, Registrar; all users authenticate here.
2. Student Registration and Profile Management	Admission applications, student registration, personal and contact data, guardians, enrollment, documents, profile corrections.	Registrar; authorized student self-service.
3. Academic Record Management	Departments, programs, courses, class offerings, schedules, faculty assignments, attendance, grades, academic history.	Department Heads, Faculty Members, Registrar.
4. Reports and Analytics Generation	Class lists, enrollment summaries, report cards, transcripts, attendance and grade reports, departmental statistics.	Registrar, Department Heads, Faculty Members, authorized administrators.

1.2 System-wide supporting capabilities

The following functions support all primary modules and should be integrated across the application:

- Audit trails and activity history
- Role-based security and privacy controls
- Notifications and attendance alerts
- Automated backup and recovery operations
- Data retention and archival controls
- System monitoring, error logging, and performance tracking

1.3 Design conventions

Convention	Recommendation
Primary keys	Use a stable internal identifier such as INT/BIGINT or UUID. Do not rely on a display number as the database primary key.
Foreign keys	Use fields named consistently, such as StudentID, ProgramID, or AcademicTermID, and enforce referential integrity.
StudentID vs StudentNumber	StudentID is the internal database key. StudentNumber is the official school identifier shown in the interface and documents.
Status fields	Use controlled values or lookup/reference tables. Avoid allowing users to type arbitrary status labels.
Dates and timestamps	Use DATE for calendar dates, TIME for time-of-day, and DATETIME/TIMESTAMP for exact events. Store consistently in the server/database timezone policy.
Money and numeric grades	Use DECIMAL where precision matters. Avoid FLOAT for grades, weighted scores, and units.
File storage	Store uploaded files in controlled storage and save a storage key/path and metadata in the database. Do not ordinarily place large file binaries in a normal row.
Passwords	Store only a strong, adaptive password hash (for example, Argon2id or bcrypt), never plaintext or reversible password text.

Convention	Recommendation
Personal data	Collect only information that has a defined institutional purpose. Restrict, log, and protect sensitive information.
Computed values	Calculate totals, ages, GPA, attendance percentages, and counts from source records. Cache only where performance requires it.

1.4 Suggested status vocabularies

Record type	Suggested values
User account	Pending Activation, Active, Inactive, Suspended, Locked
Student	Active, Inactive, On Leave, Graduated, Withdrawn, Dropped
Admission application	Pending, Under Review, Accepted, Rejected, Withdrawn
Enrollment	Pending, Verified, Enrolled, Dropped, Withdrawn, Cancelled
Document verification	Pending, Verified, Rejected, Needs Resubmission
Grade	Draft, Submitted, Under Review, Approved, Posted, Returned for Correction
Schedule	Active, Cancelled, Replaced, Completed
Report job	Queued, Processing, Completed, Failed, Expired
Backup job	Pending, Running, Completed, Failed

Configuration note These values are examples. Align them with the school's official terminology and avoid mixing a record's lifecycle status with a user's permissions.

2. Foundational Data Entities

Foundational entities are shared across modules. They prevent repeated encoding of the same person, department, program, term, or course in multiple places.

2.1 Person identity (Persons)

Recommended shared identity table for anyone who may have a student or staff record. This avoids duplicating names and contact details between Students and Staff.

Field	Suggested type	Purpose / notes
PersonID	INT / UUID	Unique internal identifier. Primary key.
FirstName	VARCHAR(100)	Person's first name.
MiddleName	VARCHAR(100)	Middle name, if applicable.
LastName	VARCHAR(100)	Surname.
Suffix	VARCHAR(20)	Jr., Sr., III, or other suffix.
PreferredName	VARCHAR(100)	Name the person prefers to be addressed by.
DateOfBirth	DATE	Birthdate, if collected for a defined purpose.
Gender	VARCHAR(30) / lookup	Only if institutionally required.
Nationality	VARCHAR(100)	Nationality, if required.
PersonalEmail	VARCHAR(255)	Personal email address.
MobileNumber	VARCHAR(30)	Primary contact number.
AlternateNumber	VARCHAR(30)	Optional secondary number.
CreatedAt	DATETIME	Date and time identity record was created.
UpdatedAt	DATETIME	Date and time identity record was last updated.

2.2 Student record (Students)

Student-specific information should be linked to the shared Person record. Student profile and academic modules should use StudentID as their primary reference.

Field	Suggested type	Purpose / notes
StudentID	INT / UUID	Unique internal student identifier. Primary key.
PersonID	INT / UUID	Foreign key to Persons; one person identity for the student.
StudentNumber	VARCHAR(30)	Official student number; unique and displayed to users.
InstitutionalEmail	VARCHAR(255)	School-issued email address.
ProfilePicture	VARCHAR(500)	Storage reference for profile image.
AddressLine	VARCHAR(255)	House number, street, building, or other address line.
Barangay	VARCHAR(100)	Barangay.
City	VARCHAR(100)	City or municipality.
Province	VARCHAR(100)	Province.
PostalCode	VARCHAR(10)	Postal code.
Country	VARCHAR(100)	Country of residence.
StudentStatus	VARCHAR(30) / lookup	Active, Inactive, On Leave, Graduated, Withdrawn, Dropped, etc.
AdmissionDate	DATE	Date admitted.

Field	Suggested type	Purpose / notes
AdmissionType	VARCHAR(30) / lookup	Freshman, Transferee, Returnee, or another institutional category.
DepartmentID	INT	Assigned department reference.
ProgramID	INT	Assigned academic program reference.
CurrentYearLevel	VARCHAR(20)	Current year level; preferably derived from the current enrollment when possible.
CreatedAt	DATETIME	Record creation timestamp.
UpdatedAt	DATETIME	Last modification timestamp.

Identifier rule StudentID and StudentNumber must be different. StudentID remains stable even if the official student number is reformatted or corrected.

2.3 Staff record (Staff)

Recommended because Department Heads, faculty, Registrar personnel, ICT Services, and Education Technology Services are system users and may need personnel details.

Field	Suggested type	Purpose / notes
StaffID	INT / UUID	Unique staff identifier. Primary key.
PersonID	INT / UUID	Foreign key to Persons.
EmployeeNumber	VARCHAR(30)	Official employee/staff number; unique where applicable.
DepartmentID	INT	Primary assigned department.
PositionTitle	VARCHAR(100)	Position or designation.
EmploymentStatus	VARCHAR(30)	Active, Inactive, On Leave, Separated, etc.
WorkEmail	VARCHAR(255)	Official work email.
WorkContactNumber	VARCHAR(30)	Official contact number, if needed.
DateHired	DATE	Employment start date, if needed.
CreatedAt	DATETIME	Record creation timestamp.
UpdatedAt	DATETIME	Last modification timestamp.

2.4 Department information (Departments)

Departments provide the organizational structure for programs, students, class offerings, faculty assignments, and departmental reports.

Field	Suggested type	Purpose / notes
DepartmentID	INT / UUID	Unique department identifier. Primary key.
DepartmentCode	VARCHAR(20)	Short code, such as IT or CCS.
DepartmentName	VARCHAR(150)	Full department name.
DepartmentDescription	TEXT	Description of the department.
DepartmentHeadStaffID	INT / UUID	Reference to the staff member serving as department head.
DepartmentEmail	VARCHAR(255)	Official department email.
ContactNumber	VARCHAR(30)	Department contact number.
OfficeLocation	VARCHAR(100)	Office location.
DepartmentStatus	VARCHAR(20)	Active or Inactive.

2.5 Academic programs (Programs)

A department may offer more than one academic program. Keep program data separate from student records.

Field	Suggested type	Purpose / notes
ProgramID	INT / UUID	Unique program identifier. Primary key.
DepartmentID	INT / UUID	Department offering the program.
ProgramCode	VARCHAR(30)	Program abbreviation, such as BSIT or BSCS.
ProgramName	VARCHAR(150)	Complete academic program name.
ProgramDescription	TEXT	Program description.
DegreeLevel	VARCHAR(30)	Undergraduate, Graduate, or other level.
ProgramDurationYears	DECIMAL(3,1) / INT	Typical program duration in years.
ProgramStatus	VARCHAR(20)	Active or Inactive.

2.6 Academic years and semesters (AcademicTerms)

An academic term is the common reference for enrollments, courses, grades, attendance reporting, and academic statistics.

Field	Suggested type	Purpose / notes
AcademicTermID	INT / UUID	Unique academic term identifier. Primary key.
AcademicYear	VARCHAR(20)	Example: 2026-2027.
Semester	VARCHAR(30)	1st Semester, 2nd Semester, Summer, Trimester, etc.
StartDate	DATE	Beginning of academic term.
EndDate	DATE	End of academic term.
EnrollmentStartDate	DATE	Beginning of enrollment period.
EnrollmentEndDate	DATE	End of enrollment period.
TermStatus	VARCHAR(20)	Upcoming, Ongoing, Completed, Archived.
IsCurrentTerm	BOOLEAN	Indicates current institution-wide term; enforce only one current term if that is the policy.

2.7 Course or subject catalogue (Courses)

A Course is the reusable subject definition. A ClassOffering is a specific instance of that course in a semester and section. If prerequisite checking is required, add an optional CoursePrerequisites junction table linking CourseID to PrerequisiteCourseID, with MinimumGrade and RequirementType.

Field	Suggested type	Purpose / notes
CourseID	INT / UUID	Unique course identifier. Primary key.
CourseCode	VARCHAR(30)	Course code, such as IT101.
CourseName	VARCHAR(150)	Full subject name.
CourseDescription	TEXT	Course description.
DepartmentID	INT / UUID	Department responsible for the course.
Units	DECIMAL(4,2)	Academic units.
LectureHours	DECIMAL(4,2)	Lecture hours.
LaboratoryHours	DECIMAL(4,2)	Laboratory hours.
CourseType	VARCHAR(30)	Major, Minor, General Education, Elective, etc.
IsActive	BOOLEAN	Indicates whether the course is currently offered.
CourseStatus	VARCHAR(20)	Active or Inactive.

3. Module 1 - User Authentication and Access Control

This module authenticates users and determines what each user can view, create, modify, approve, export, or administer. The supplied blue login mockup is a suitable visual starting point; account and permission logic should be designed independently from the visual layout.

3.1 Login interface

Login form fields

Field	Suggested type	Purpose / notes
UsernameOrEmail	VARCHAR(255)	Username or registered email supplied at login.
Password	Input / secret	Password supplied by the user. Never store this raw value.
RememberMe	BOOLEAN	Optional persistent session on a trusted device, subject to policy.
ForgotPassword	Action / link	Starts the password recovery flow.
MFAChallengeCode	VARCHAR(20)	Optional one-time verification code for multi-factor authentication.

3.2 User account management (Users)

ICT Services should maintain system accounts and account security. A separate person, student, or staff record should hold identity and school information; Users should hold authentication-related information.

Field	Suggested type	Purpose / notes
UserID	INT / UUID	Unique account identifier. Primary key.
PersonID	INT / UUID	Associated person record; optional for special service accounts.
Username	VARCHAR(100)	Unique login name.
Email	VARCHAR(255)	Registered account email; unique according to policy.
PasswordHash	VARCHAR(255)	Secure adaptive hash; never plaintext.
AccountStatus	VARCHAR(30)	Pending Activation, Active, Inactive, Suspended, Locked.
EmailVerified	BOOLEAN	Whether email verification has been completed.
MFAEnabled	BOOLEAN	Whether multi-factor authentication is enabled.
FailedLoginAttempts	INT	Consecutive unsuccessful login attempts.
LockoutUntil	DATETIME	Temporary lockout end time.
LastLoginAt	DATETIME	Most recent successful login.
PasswordChangedAt	DATETIME	Most recent password change.
MustChangePassword	BOOLEAN	Requires password update at next login.
CreatedAt	DATETIME	Account creation timestamp.
CreatedBy	INT / UUID	UserID of creator, if created by an administrator.
UpdatedAt	DATETIME	Last modification timestamp.

Account lifecycle Deactivating a user account should not delete the associated student or staff record. Account access and institutional identity records have separate lifecycles.

3.3 Roles (Roles)

Role fields

Field	Suggested type	Purpose / notes
RoleID	INT / UUID	Unique role identifier. Primary key.
RoleName	VARCHAR(100)	Human-readable role name.
RoleCode	VARCHAR(30)	Stable code, such as REGISTRAR or FACULTY.
RoleDescription	TEXT	Responsibilities and scope of the role.
RoleLevel	INT	Optional hierarchy/ranking. Avoid using this as the only permission mechanism.
RoleStatus	VARCHAR(20)	Active or Inactive.
CreatedAt	DATETIME	Creation timestamp.

Initial roles from the scenario:

Role code	Role name
ICT_ADMIN	Information and Communication Technology Services
EDTECH	Education Technology Services
DEPT_HEAD	Department Head
FACULTY	Faculty Member
REGISTRAR	Registrar

Optional sixth role - Student The scenario lists five staff user groups, but the supplied blue mockup includes a student profile/course view. If students are expected to log in, add a STUDENT role with tightly limited self-service permissions: view own profile, enrollment, courses, grades, attendance, and documents; submit profile correction requests; and receive notifications. Students should not edit official records directly.

3.4 Permissions (Permissions)

Permission fields

Field	Suggested type	Purpose / notes
PermissionID	INT / UUID	Unique permission identifier.
PermissionName	VARCHAR(100)	Readable permission name.
PermissionCode	VARCHAR(100)	Unique code, e.g. STUDENT_EDIT or GRADE_VIEW.
ModuleName	VARCHAR(100)	Module to which the permission applies.
ActionType	VARCHAR(30)	Create, Read, Update, Delete, Approve, Post, Export, Manage, etc.
PermissionDescription	TEXT	What the permission allows.

3.5 Role and user assignments

UserRoles junction table

Field	Suggested type	Purpose / notes
UserRoleID	INT / UUID	Unique user-to-role assignment identifier.
UserID	INT / UUID	User receiving the role.
RoleID	INT / UUID	Assigned role.

Field	Suggested type	Purpose / notes
DepartmentID	INT / UUID	Optional scope: department to which this role applies.
EffectiveFrom	DATE	Date the assignment becomes effective.
EffectiveUntil	DATE	Optional end date.
AssignedAt	DATETIME	When the role was assigned.
AssignedBy	INT / UUID	Administrator who assigned it.
AssignmentStatus	VARCHAR(20)	Active, Expired, Revoked.

RolePermissions junction table

Field	Suggested type	Purpose / notes
RolePermissionID	INT / UUID	Unique role-permission assignment.
RoleID	INT / UUID	Role receiving permission.
PermissionID	INT / UUID	Permission being granted.
GrantedAt	DATETIME	When permission was granted.
GrantedBy	INT / UUID	Administrator who granted it.

Use a many-to-many relationship between Users and Roles through UserRoles, and between Roles and Permissions through RolePermissions. A simpler one-role-per-user design can place RoleID in Users, but the junction-table approach is more flexible and is recommended.

3.6 Password recovery (PasswordResetRequests)

Field	Suggested type	Purpose / notes
ResetID	INT / UUID	Unique recovery request identifier.
UserID	INT / UUID	Account requesting recovery.
ResetTokenHash	VARCHAR(255)	Securely stored token hash; do not store a usable token in plain text.
RequestDate	DATETIME	Request timestamp.
ExpirationDate	DATETIME	Token expiry.
IsUsed	BOOLEAN	Whether token has been consumed.
DateUsed	DATETIME	When the reset was completed.

3.7 Authentication and account rules

- Require unique usernames and enforce email uniqueness where the school uses email as a login identifier.
- Use secure password hashing, rate limits, account lockout or step-up verification, and protected sessions.
- Expire password-reset tokens and make them single-use.
- Apply permissions in the backend/API for every protected action; interface hiding is not access control.
- Log successful and failed authentication events without recording passwords or authentication secrets.
- Review and revoke roles when personnel change positions, leave a department, or separate from the institution.

4. Module 2 - Student Registration and Profile Management

This module supports the Registrar in registering new students, maintaining personal and contact information, processing enrollment, handling profile corrections, and storing admission requirements. It also provides controlled self-service where students are allowed to view or request changes to their records.

4.1 Admission applications (AdmissionApplications)

Keep the application separate from the permanent student record because applicants may be rejected, withdraw, or never enroll. Convert accepted applicants into student records according to the admissions process.

Field	Suggested type	Purpose / notes
ApplicationID	INT / UUID	Unique admission application identifier.
ApplicationNumber	VARCHAR(30)	Official application reference; unique.
PersonID	INT / UUID	Applicant identity record.
StudentID	INT / UUID	Linked student after conversion/acceptance, if applicable.
ApplicationDate	DATE	Date submitted.
AdmissionType	VARCHAR(30)	Freshman, Transferee, Returnee, etc.
AppliedProgramID	INT / UUID	Program applied for.
IntendedAcademicTermID	INT / UUID	Academic term for intended entry.
ApplicationStatus	VARCHAR(30)	Pending, Under Review, Accepted, Rejected, Withdrawn.
AdmissionDecisionDate	DATE	Date of decision.
VerifiedBy	INT / UUID	Staff/User who verified the application.
VerificationDate	DATETIME	Date and time of verification.
AdmissionRemarks	TEXT	Additional decision notes.

4.2 Student registration - personal information

Field ownership If the shared Persons table is used, names, date of birth, personal email, and primary contact details should be stored there; Students should hold student-specific identifiers, academic status, program/department links, and school-specific data. The form below describes the complete registration experience, not a requirement to duplicate fields across tables.

A. Personal information

Field	Suggested type	Purpose / notes
StudentID	INT / UUID	Internal student identifier; linked to Students.
StudentNumber	VARCHAR(30)	Official student number.
FirstName	VARCHAR(100)	First name; normally sourced from Persons.
MiddleName	VARCHAR(100)	Middle name.
LastName	VARCHAR(100)	Surname.
Suffix	VARCHAR(20)	Name suffix.
PreferredName	VARCHAR(100)	Preferred display name.
DateOfBirth	DATE	Date of birth.

Field	Suggested type	Purpose / notes
Gender	VARCHAR(30) / lookup	Only if collected under institutional requirements.
Nationality	VARCHAR(100)	Nationality if required.
CivilStatus	VARCHAR(30)	Optional; collect only where needed.
PlaceOfBirth	VARCHAR(150)	Optional place of birth.
ProfilePicture	VARCHAR(500)	Secure image storage reference.

B. Contact information

Field	Suggested type	Purpose / notes
PersonalEmail	VARCHAR(255)	Student personal email.
InstitutionalEmail	VARCHAR(255)	School-issued email.
MobileNumber	VARCHAR(30)	Primary student contact.
AlternateNumber	VARCHAR(30)	Secondary contact.
AddressLine	VARCHAR(255)	House/building/street address.
Barangay	VARCHAR(100)	Barangay.
City	VARCHAR(100)	City or municipality.
Province	VARCHAR(100)	Province.
PostalCode	VARCHAR(10)	Postal code.
Country	VARCHAR(100)	Country of residence.

C. Parent / guardian information (Guardians)

Field	Suggested type	Purpose / notes
GuardianID	INT / UUID	Unique guardian record identifier.
StudentID	INT / UUID	Student associated with the guardian.
GuardianFirstName	VARCHAR(100)	Guardian first name.
GuardianMiddleName	VARCHAR(100)	Guardian middle name.
GuardianLastName	VARCHAR(100)	Guardian surname.
Relationship	VARCHAR(50)	Mother, Father, Guardian, etc.
ContactNumber	VARCHAR(30)	Primary guardian contact.
EmailAddress	VARCHAR(255)	Guardian email.
Address	VARCHAR(255)	Guardian address; may be same as student address.
IsPrimaryContact	BOOLEAN	Indicates primary contact.
IsEmergencyContact	BOOLEAN	Indicates emergency-contact designation.

D. Emergency contacts (EmergencyContacts)

Field	Suggested type	Purpose / notes
EmergencyContactID	INT / UUID	Unique emergency-contact identifier.
StudentID	INT / UUID	Associated student.
ContactName	VARCHAR(150)	Full name of emergency contact.
Relationship	VARCHAR(50)	Relationship to the student.
ContactNumber	VARCHAR(30)	Primary emergency number.
AlternateNumber	VARCHAR(30)	Secondary number.
Address	VARCHAR(255)	Address, if needed.

E. Educational background

Field	Suggested type	Purpose / notes
EducationalBackgroundID	INT / UUID	Unique educational-background record.
StudentID	INT / UUID	Associated student.
PreviousSchool	VARCHAR(200)	Previously attended institution.
PreviousSchoolAddress	VARCHAR(255)	Previous school address.
PreviousSchoolLevel	VARCHAR(50)	Senior High School, College, etc.
StrandOrTrack	VARCHAR(100)	Previous strand/track, if applicable.
YearGraduated	INT / YEAR	Year graduated.
PreviousProgram	VARCHAR(150)	Previous college program, if applicable.
IsTransferee	BOOLEAN	Indicates transferee classification.
TransferCredentialsVerified	BOOLEAN	Indicates transfer credential verification.

4.3 Student profile management

The supplied blue mockup shows a profile-editing interface with General, User Picture, and Contact Information areas. These can be presented as tabs or sections in one profile page.

Optional profile control fields

Field	Suggested type	Purpose / notes
ProfileID	INT / UUID	Unique profile record, if profile data is maintained separately from Students/Persons.
StudentID	INT / UUID	Associated student.
LastProfileUpdate	DATETIME	Date and time of last profile update.
LastUpdatedBy	INT / UUID	User who last modified the profile.
ProfileCompletionStatus	VARCHAR(30)	Incomplete, Complete, or other completeness state.
ProfileVerificationStatus	VARCHAR(30)	Pending, Verified, Needs Correction.

4.4 Profile correction requests (ProfileCorrectionRequests)

Students may request corrections, while the Registrar or another authorized reviewer approves and applies changes. Keep a history rather than silently overwriting official values.

Field	Suggested type	Purpose / notes
CorrectionRequestID	INT / UUID	Unique correction request identifier.
StudentID	INT / UUID	Student submitting request.
FieldToChange	VARCHAR(100)	Field or attribute to be corrected.
ExistingValue	TEXT	Current official value; protect sensitive values.
RequestedValue	TEXT	Proposed replacement value.
RequestReason	TEXT	Reason for correction.
SupportingDocumentID	INT / UUID	Supporting evidence reference, if applicable.
RequestDate	DATETIME	Request timestamp.
RequestStatus	VARCHAR(30)	Pending, Approved, Rejected, Cancelled.
ReviewedBy	INT / UUID	Authorized reviewer.
ReviewDate	DATETIME	Date reviewed.
ReviewRemarks	TEXT	Decision explanation.

Field	Suggested type	Purpose / notes
AppliedAt	DATETIME	When approved correction was applied.

4.5 Student enrollment (StudentEnrollments)

Enrollment represents a student's overall registration for one academic term. It should not be confused with the individual course or section enrollments that make up the student's schedule.

Field	Suggested type	Purpose / notes
EnrollmentID	INT / UUID	Unique term enrollment identifier.
StudentID	INT / UUID	Student being enrolled.
AcademicTermID	INT / UUID	Academic year and semester.
ProgramID	INT / UUID	Program for the term.
DepartmentID	INT / UUID	Department for the term.
YearLevel	VARCHAR(20)	Year level during the term.
EnrollmentDate	DATETIME	Enrollment processing date.
EnrollmentStatus	VARCHAR(30)	Pending, Verified, Enrolled, Dropped, Withdrawn, Cancelled.
TotalRegisteredUnits	DECIMAL(5,2)	Sum of active enrolled course units; preferably calculated.
EnrollmentReferenceNumber	VARCHAR(50)	Official enrollment reference.
ProcessedBy	INT / UUID	Staff member who processed the enrollment.
ApprovedBy	INT / UUID	Authorized approver.
ApprovalDate	DATETIME	Approval timestamp.
EnrollmentRemarks	TEXT	Additional notes.

4.6 Individual course enrollment (ClassEnrollments)

A student can have one term enrollment and several individual course enrollments. This junction table connects a student's term registration to a specific ClassOffering.

Field	Suggested type	Purpose / notes
ClassEnrollmentID	INT / UUID	Unique student-to-class enrollment identifier.
EnrollmentID	INT / UUID	Associated overall term enrollment.
StudentID	INT / UUID	Student identifier; useful for querying and validation.
ClassOfferingID	INT / UUID	Specific course/section offering.
DateAdded	DATETIME	When the class was added.
EnrollmentStatus	VARCHAR(30)	Enrolled, Dropped, Withdrawn, Completed.
DateDropped	DATETIME	When dropped, if applicable.
DropReason	TEXT	Reason for dropping, if applicable.
AddedBy	INT / UUID	User who added the student to the class.

Relationship rule One StudentEnrollment can be linked to many ClassEnrollments. Enforce uniqueness for an active StudentID + ClassOfferingID combination and prevent duplicate enrollment in the same class.

4.7 Document type catalogue (DocumentTypes)

Field	Suggested type	Purpose / notes
DocumentTypeID	INT / UUID	Unique document-type identifier.
DocumentName	VARCHAR(150)	Name of the required document.
DocumentDescription	TEXT	Description of the requirement.
ApplicableAdmissionType	VARCHAR(50)	Freshman, Transferee, Returnee, All, or configurable rule.
IsRequired	BOOLEAN	Whether submission is mandatory.
AllowedFileTypes	VARCHAR(255) / JSON	Accepted formats/extensions.
MaximumFileSize	BIGINT	Maximum permitted size in bytes.
IsActive	BOOLEAN	Whether requirement is currently active.

Potential examples: PSA birth certificate, Form 138/report card, good moral certificate, certificate of transfer credentials, transcript of records, identification photograph, and other documents required by the institution.

4.8 Student documents (StudentDocuments)

Field	Suggested type	Purpose / notes
StudentDocumentID	INT / UUID	Unique uploaded document identifier.
StudentID	INT / UUID	Student who owns the document.
DocumentTypeID	INT / UUID	Type of document.
FileName	VARCHAR(255)	Original uploaded file name.
StoragePath	VARCHAR(500)	Secure storage key or file reference.
FileType	VARCHAR(50)	PDF, JPG, PNG, etc.
FileSize	BIGINT	Size in bytes.
FileHash	VARCHAR(255)	Integrity/checksum value.
VersionNumber	INT	Document version.
UploadDate	DATETIME	Upload timestamp.
UploadedBy	INT / UUID	User who uploaded the file.
VerificationStatus	VARCHAR(30)	Pending, Verified, Rejected, Needs Resubmission.
VerifiedBy	INT / UUID	Reviewer.
VerificationDate	DATETIME	Verification timestamp.
RejectionReason	TEXT	Reason for rejection.
ExpirationDate	DATE	Expiration, if applicable.
RetentionUntil	DATE	Retention end date, if applicable.

4.9 Document verification history (DocumentVerifications)

Field	Suggested type	Purpose / notes
VerificationID	INT / UUID	Unique verification event identifier.
StudentDocumentID	INT / UUID	Document reviewed.
VerifiedBy	INT / UUID	Reviewer.
VerificationStatus	VARCHAR(30)	Verification result.
VerificationRemarks	TEXT	Reviewer comments.
VerificationDate	DATETIME	Date and time of review.

5. Module 3 - Academic Record Management

This module manages the academic structures and transactions that occur after admission: course offerings, class sections, faculty assignments, schedules, attendance, assessments, grades, and consolidated academic history.

5.1 Class offerings (ClassOfferings)

A ClassOffering is a specific scheduled instance of a Course for a particular academic term, department/program, and section.

Field	Suggested type	Purpose / notes
ClassOfferingID	INT / UUID	Unique class offering identifier.
CourseID	INT / UUID	Course/subject being offered.
AcademicTermID	INT / UUID	Academic year and semester.
DepartmentID	INT / UUID	Responsible department.
ProgramID	INT / UUID	Associated program, where applicable.
SectionCode	VARCHAR(30)	Section designation, such as 3B or IT-301.
YearLevel	VARCHAR(20)	Intended year level.
ClassCapacity	INT	Maximum number of students.
CurrentClassSize	INT / derived	Number of active students; calculate from enrollments rather than manually maintaining where possible.
ClassType	VARCHAR(30)	Lecture, Laboratory, Combined.
ClassModality	VARCHAR(30)	Face-to-Face, Online, Hybrid, if applicable.
ClassStatus	VARCHAR(20)	Open, Full, Closed, Cancelled, Completed.
CreatedBy	INT / UUID	User who created offering.
CreatedAt	DATETIME	Creation timestamp.

5.2 Faculty assignments (FacultyAssignments)

Field	Suggested type	Purpose / notes
FacultyAssignmentID	INT / UUID	Unique faculty assignment identifier.
ClassOfferingID	INT / UUID	Class/section assigned.
FacultyStaffID	INT / UUID	Assigned faculty staff record.
AssignmentRole	VARCHAR(30)	Main Instructor, Assistant, Co-Instructor, etc.
AssignedBy	INT / UUID	Department Head or authorized assigner.
AssignmentDate	DATETIME	Assignment timestamp.
AssignmentStatus	VARCHAR(20)	Active, Completed, Cancelled.

5.3 Class schedules (ClassSchedules)

Field	Suggested type	Purpose / notes
ScheduleID	INT / UUID	Unique schedule identifier.
ClassOfferingID	INT / UUID	Associated class offering.
DayOfWeek	VARCHAR(15)	Monday, Tuesday, etc.
StartTime	TIME	Scheduled start.
EndTime	TIME	Scheduled end.
RoomID	INT / UUID	Assigned room.

Field	Suggested type	Purpose / notes
Building	VARCHAR(100)	Building location; can be derived from Room if normalized.
ScheduleType	VARCHAR(30)	Lecture, Laboratory, etc.
EffectiveFrom	DATE	Date schedule becomes effective.
EffectiveUntil	DATE	Date schedule ends, if applicable.
ScheduleStatus	VARCHAR(20)	Active, Cancelled, Replaced.
LastModifiedBy	INT / UUID	User who last changed schedule.
LastModifiedAt	DATETIME	Modification timestamp.

5.4 Room catalogue (Rooms)

Field	Suggested type	Purpose / notes
RoomID	INT / UUID	Unique room identifier.
RoomCode	VARCHAR(30)	Room designation, e.g. ITLAB-01.
RoomName	VARCHAR(100)	Full room name.
Building	VARCHAR(100)	Building name.
Floor	VARCHAR(30)	Floor designation.
RoomCapacity	INT	Maximum occupancy.
RoomType	VARCHAR(30)	Classroom, Laboratory, Conference Room, etc.
RoomStatus	VARCHAR(30)	Available, Unavailable, Under Maintenance.

5.5 Attendance sessions (AttendanceSessions)

An attendance session represents one meeting of a class. Faculty should create or open the session for the applicable class date and then record one attendance result per enrolled student.

Field	Suggested type	Purpose / notes
AttendanceSessionID	INT / UUID	Unique class-meeting identifier.
ClassOfferingID	INT / UUID	Class where attendance is taken.
ScheduleID	INT / UUID	Corresponding schedule, if applicable.
AttendanceDate	DATE	Class date.
SessionNumber	INT	Meeting/session number, if used.
StartTime	TIME	Scheduled start time.
EndTime	TIME	Scheduled end time.
AttendanceDeadline	TIME	Attendance cutoff, if applicable.
RecordedBy	INT / UUID	Faculty or authorized user.
RecordedAt	DATETIME	Timestamp attendance was recorded/submitted.
SessionRemarks	TEXT	Additional notes.
SessionStatus	VARCHAR(20)	Open, Submitted, Locked.

5.6 Student attendance records (AttendanceRecords)

Field	Suggested type	Purpose / notes
AttendanceID	INT / UUID	Unique student attendance record.
AttendanceSessionID	INT / UUID	Associated class meeting.

Field	Suggested type	Purpose / notes
StudentID	INT / UUID	Student whose attendance is recorded.
ClassEnrollmentID	INT / UUID	Associated course enrollment.
AttendanceStatus	VARCHAR(20)	Present, Absent, Late, Excused, Left Early.
TimeIn	TIME	Actual arrival time.
TimeOut	TIME	Actual departure time, if recorded.
MinutesLate	INT	Late minutes; calculate from scheduled start and policy/grace period.
ReasonForAbsence	TEXT	Absence explanation, if provided.
Remarks	TEXT	Additional attendance notes.
RecordedBy	INT / UUID	User who entered record.
RecordedAt	DATETIME	Entry timestamp.
LastModifiedBy	INT / UUID	User who last changed record.
LastModifiedAt	DATETIME	Modification timestamp.

Uniqueness constraint Enforce a unique AttendanceSessionID + StudentID combination for each attendance record. Attendance should reference a valid enrolled student and a valid class meeting.

5.7 Attendance alerts (AttendanceAlerts)

Field	Suggested type	Purpose / notes
AlertID	INT / UUID	Unique alert identifier.
StudentID	INT / UUID	Student associated with alert.
AttendanceID	INT / UUID	Attendance record that triggered alert.
RecipientName	VARCHAR(150)	Guardian/parent recipient name, if applicable.
RecipientContact	VARCHAR(255)	Email address or phone number; restrict access.
AlertType	VARCHAR(30)	Absence, Tardiness, Attendance Threshold.
AlertMessage	TEXT	Notification content.
NotificationChannel	VARCHAR(20)	Email, SMS, Portal.
DateTriggered	DATETIME	When generated.
DateSent	DATETIME	When sent.
DeliveryStatus	VARCHAR(30)	Pending, Sent, Failed, Delivered.

5.8 Attendance alert settings (AttendanceAlertSettings)

Field	Suggested type	Purpose / notes
AlertSettingID	INT / UUID	Unique alert configuration identifier.
MaximumAllowedAbsences	INT	Absence threshold before an alert.
TardinessThreshold	INT	Number of late arrivals before notification.
NotifyGuardian	BOOLEAN	Whether guardian notifications are enabled.
NotifyFaculty	BOOLEAN	Whether faculty notifications are enabled.
NotifyRegistrar	BOOLEAN	Whether Registrar notifications are enabled, if policy requires.
IsEnabled	BOOLEAN	Enables or disables this rule.
EffectiveFrom	DATE	Rule effective date.

Field	Suggested type	Purpose / notes
EffectiveUntil	DATE	Optional end date.

Guardian alerts are a proposed extension of the requirement to generate absence alerts for teachers or parents. The institution must decide which recipients are appropriate, what thresholds apply, and whether notifications use portal messages, email, or SMS.

5.9 Grading periods (GradingPeriods)

Field	Suggested type	Purpose / notes
GradingPeriodID	INT / UUID	Unique grading period identifier.
AcademicTermID	INT / UUID	Associated academic term.
PeriodName	VARCHAR(50)	Prelim, Midterm, Final, etc.
PeriodStartDate	DATE	Period start.
PeriodEndDate	DATE	Period end.
SubmissionDeadline	DATETIME	Grade submission cutoff.
PeriodStatus	VARCHAR(20)	Upcoming, Open, Closed, Finalized.

5.10 Assessments (Assessments)

Field	Suggested type	Purpose / notes
AssessmentID	INT / UUID	Unique assessment identifier.
ClassOfferingID	INT / UUID	Class offering where assessment applies.
GradingPeriodID	INT / UUID	Associated grading period.
AssessmentName	VARCHAR(150)	Quiz 1, Midterm Examination, Project, etc.
AssessmentType	VARCHAR(50)	Quiz, Examination, Laboratory, Project, Assignment, etc.
MaximumScore	DECIMAL(8,2)	Highest possible score.
WeightPercentage	DECIMAL(5,2)	Assessment contribution to grading period/grade.
AssessmentDate	DATE	Assessment date.
CreatedBy	INT / UUID	Faculty member who created assessment.
DateCreated	DATETIME	Creation timestamp.
AssessmentStatus	VARCHAR(20)	Active, Completed, Cancelled.

5.11 Student assessment scores (AssessmentScores)

MaximumScore is normally read from Assessments rather than duplicated in every AssessmentScores row. If a snapshot is required to preserve historical scoring after an assessment definition changes, explicitly label it as a snapshot field and maintain it consistently.

Field	Suggested type	Purpose / notes
AssessmentScoreID	INT / UUID	Unique score record.
AssessmentID	INT / UUID	Associated assessment.
StudentID	INT / UUID	Student receiving score.
RawScore	DECIMAL(8,2)	Points earned.
PercentageScore	DECIMAL(5,2)	Computed percentage based on RawScore and Assessment.MaximumScore.
Remarks	TEXT	Optional faculty comment.
EncodedBy	INT / UUID	Faculty user who entered score.

Field	Suggested type	Purpose / notes
EncodedAt	DATETIME	Entry timestamp.
UpdatedBy	INT / UUID	User who last changed score.
UpdatedAt	DATETIME	Last modification timestamp.

5.12 Official student grades (StudentGrades)

Field	Suggested type	Purpose / notes
GradeID	INT / UUID	Unique official/period grade identifier.
StudentID	INT / UUID	Student receiving grade.
ClassEnrollmentID	INT / UUID	Course enrollment associated with the grade.
AcademicTermID	INT / UUID	Academic term.
GradingPeriodID	INT / UUID	Grading period, if storing period-level grades.
PreliminaryGrade	DECIMAL(5,2)	Preliminary grade, if used.
MidtermGrade	DECIMAL(5,2)	Midterm grade, if used.
FinalGrade	DECIMAL(5,2)	Final numerical grade.
LetterGrade	VARCHAR(5)	Letter equivalent, if used.
GradePoint	DECIMAL(4,2)	Grade point equivalent, if used.
Remarks	VARCHAR(50)	Passed, Failed, Incomplete, Withdrawn, etc.
IsOfficial	BOOLEAN	Whether record is official.
GradeStatus	VARCHAR(30)	Draft, Submitted, Under Review, Approved, Posted.
EncodedBy	INT / UUID	Faculty member who encoded grade.
SubmittedAt	DATETIME	Submission timestamp.
ReviewedBy	INT / UUID	Reviewer, such as Department Head.
ReviewedAt	DATETIME	Review timestamp.
ApprovedBy	INT / UUID	Authorized approver.
ApprovedAt	DATETIME	Approval timestamp.
PostedBy	INT / UUID	User who finalized/post grade.
PostedAt	DATETIME	Official posting timestamp.

Grading model Support the institution's actual grading scheme. Do not hardcode a fixed set of Prelim/Midterm/Final columns if the school may use different periods. A flexible GradingPeriods table and period-grade records are more adaptable.

5.13 Grade scale (GradeScales)

Field	Suggested type	Purpose / notes
GradeScaleID	INT / UUID	Unique grade-scale row.
GradeEquivalent	DECIMAL(5,2) / VARCHAR(10)	Numerical or letter grade equivalent.
MinimumPercentage	DECIMAL(5,2)	Lowest percentage in range.
MaximumPercentage	DECIMAL(5,2)	Highest percentage in range.
GradePoint	DECIMAL(4,2)	Corresponding grade point, if applicable.
VerbalEquivalent	VARCHAR(50)	Excellent, Very Good, Satisfactory, etc.
IsPassing	BOOLEAN	Indicates passing grade.

Field	Suggested type	Purpose / notes
AcademicTermID	INT / UUID	Applicable term, if grading policy varies by term.

5.14 Academic record summary (StudentAcademicRecords)

Official transcript content should normally be assembled from approved course enrollments and grades. A summary table may cache term-level values for performance, but the underlying transactions remain the source of truth.

Field	Suggested type	Purpose / notes
AcademicRecordID	INT / UUID	Unique term-summary identifier.
StudentID	INT / UUID	Associated student.
AcademicTermID	INT / UUID	Academic term.
ProgramID	INT / UUID	Program during the term.
YearLevel	VARCHAR(20)	Year level during the term.
TotalUnitsEnrolled	DECIMAL(5,2)	Total enrolled units.
TotalUnitsCompleted	DECIMAL(5,2)	Successfully completed units.
TotalUnitsFailed	DECIMAL(5,2)	Units associated with failed courses.
TermGPA	DECIMAL(4,2)	Term GPA, if applicable.
CumulativeGPA	DECIMAL(4,2)	Cumulative GPA, if applicable.
AcademicStanding	VARCHAR(50)	Good Standing, Probation, etc., based on school rules.
RecordStatus	VARCHAR(20)	Draft, Verified, Official.
LastVerifiedBy	INT / UUID	Person who verified record.
LastVerifiedAt	DATETIME	Verification timestamp.

5.15 Suggested academic workflow

1. Department Head or authorized scheduler creates ClassOfferings and assigns faculty, schedules, and rooms.
2. Registrar or authorized enrollment personnel creates the student's term StudentEnrollment and the individual ClassEnrollments.
3. Faculty accesses only their assigned class offerings and enrolled class lists.
4. Faculty records AttendanceSessions and AttendanceRecords for each meeting.
5. Faculty defines assessments, records AssessmentScores, and submits period/final grades.
6. Department Head reviews grades or academic progress where the school requires departmental approval.
7. Registrar verifies and finalizes official records according to institutional policy.
8. Approved records become available to authorized student self-service and report generation.

6. Module 4 - Reports and Analytics Generation

This module retrieves information from the other modules to generate printable summaries and exports. A report should not become a separate source of student or grade information; it should be a controlled view of existing records.

6.1 Report catalogue (ReportDefinitions)

Field	Suggested type	Purpose / notes
ReportDefinitionID	INT / UUID	Unique report template/definition identifier.
ReportCode	VARCHAR(30)	Short report code.
ReportName	VARCHAR(150)	Report name.
ReportType	VARCHAR(50)	Class List, Transcript, Enrollment Summary, etc.
ReportDescription	TEXT	Purpose and content of report.
RequiredPermissionCode	VARCHAR(100)	Permission required to generate/view report.
DefaultFormat	VARCHAR(20)	PDF, XLSX, CSV, etc.
IsActive	BOOLEAN	Whether template is available.

6.2 Generated report jobs (GeneratedReports)

Field	Suggested type	Purpose / notes
ReportID	INT / UUID	Unique report generation record.
ReportDefinitionID	INT / UUID	Report template used.
ReportCode	VARCHAR(30)	Code snapshot, if useful for historical traceability.
ReportName	VARCHAR(150)	Name snapshot.
ReportType	VARCHAR(50)	Report category.
GeneratedBy	INT / UUID	User requesting report.
DateGenerated	DATETIME	Generation timestamp.
AcademicTermID	INT / UUID	Term covered, if applicable.
DepartmentID	INT / UUID	Department filter, if applicable.
ProgramID	INT / UUID	Program filter, if applicable.
ReportFormat	VARCHAR(20)	PDF, XLSX, CSV.
ReportFilePath	VARCHAR(500)	Secure generated-file reference.
ReportStatus	VARCHAR(30)	Queued, Processing, Completed, Failed, Expired.
ExpirationDate	DATETIME	When temporary exported file expires, if applicable.

6.3 Report filter parameters

These are the filter values the interface should support. They may be stored as typed parameters or a validated JSON structure rather than requiring a separate physical column for every possible filter in every report.

Field	Suggested type	Purpose / notes
AcademicYear	VARCHAR(20)	Filter by academic year.
Semester	VARCHAR(30)	Filter by semester.

Field	Suggested type	Purpose / notes
DepartmentID	INT / UUID	Filter by department.
ProgramID	INT / UUID	Filter by academic program.
YearLevel	VARCHAR(20)	Filter by year level.
CourseID	INT / UUID	Filter by subject/course.
ClassOfferingID	INT / UUID	Filter by section/class offering.
StudentID	INT / UUID	Filter to one student, subject to permissions.
StudentStatus	VARCHAR(30)	Filter by student status.
DateFrom	DATE	Beginning of reporting period.
DateTo	DATE	End of reporting period.
FiltersApplied	JSON / TEXT	Stored report parameters used for traceability.

6.4 Report history (ReportHistory)

Field	Suggested type	Purpose / notes
ReportHistoryID	INT / UUID	Unique report history identifier.
UserID	INT / UUID	User who generated or accessed report.
ReportID	INT / UUID	Generated report job.
DateGenerated	DATETIME	Generation timestamp.
FiltersApplied	JSON / TEXT	Parameters used.
NumberOfRecords	INT	Number of records included.
ExportFormat	VARCHAR(20)	PDF, XLSX, CSV.
GenerationStatus	VARCHAR(30)	Successful or Failed.
DownloadedAt	DATETIME	Download timestamp, if tracked.

6.5 Recommended report catalogue

Category	Report	Typical contents
Student	Student Master List	Student number, full name, department, program, year level, status.
Student	Student Profile Report	Personal and contact details, guardian, admission details; permission-restricted.
Student	Student Demographic Report	Approved aggregate demographic categories and counts.
Student	Student Admission Report	Applications, admission type, decision status, admission period.
Student	Student Enrollment Report	Students enrolled in selected academic term.
Student	Student Status Report	Active, inactive, on leave, graduated, withdrawn, dropped.
Academic	Student Class List	Student number, student name, course, section, faculty.
Academic	Student Grade Report	Grades per course and grading period.
Academic	Student Report Card	Student details, subjects, grades, units, remarks.
Academic	Official Transcript of Records	Approved academic history, subjects, units, grades, academic standing.
Academic	Academic Performance Report	Grade distribution, passing/failing rates, average grades.

Category	Report	Typical contents
Academic	Academic Standing Report	Students classified by the institution's standing rules.
Academic	Course Completion Report	Completed, failed, withdrawn, and outstanding courses.
Attendance	Daily Attendance Report	Attendance for one class/date.
Attendance	Student Attendance Summary	Present, absent, late, excused totals for a student.
Attendance	Class Attendance Report	Attendance results for a complete section.
Attendance	Student Absence Report	Students with accumulated absences.
Attendance	Tardiness Report	Students with repeated late arrivals.
Attendance	Attendance Percentage Report	Attendance percentage per student or class.
Department	Department Student List	All students in a selected department.
Department	Department Enrollment Summary	Enrollment by program, year level, and term.
Department	Department Academic Performance	Average grades, pass rates, and grade distributions.
Department	Faculty Teaching Assignment Report	Faculty member, course, section, and enrollment count.
Department	Department Class Schedule	Class schedules and room assignments.
Department	Department Student Status Report	Student status distribution by department.

6.6 Report security and output

- Check report permissions and row-level scope before retrieving data. For example, Faculty should generally be limited to their assigned classes.
- Protect generated transcripts, profiles, demographic exports, and other confidential documents using authenticated download links and appropriate expiry.
- Log report generation and, where appropriate, downloads and exports in ReportHistory and AuditLogs.
- Provide preview, print, PDF export, spreadsheet export, and CSV export only when appropriate for the report type.
- Avoid including unnecessary personal information in aggregate analytics. Use de-identified or aggregated data where individual-level detail is not needed.

7. System-wide Supporting Components

These components implement the non-functional requirements in the scenario and provide shared reliability, accountability, and security capabilities.

7.1 Audit trails (AuditLogs)

Field	Suggested type	Purpose / notes
AuditLogID	INT / UUID	Unique audit event identifier.
UserID	INT / UUID	User who performed action; nullable for system events.
UsernameSnapshot	VARCHAR(100)	Optional historical username snapshot.
ModuleName	VARCHAR(100)	Module where activity occurred.
EntityName	VARCHAR(100)	Affected entity/table name.
RecordID	VARCHAR(100)	Identifier of affected record.
ActionType	VARCHAR(30)	CREATE, READ, UPDATE, DELETE, LOGIN, LOGOUT, APPROVE, EXPORT, etc.
PreviousValue	JSON / TEXT	Prior value for relevant changed fields; minimize sensitive content.
NewValue	JSON / TEXT	New value for relevant changed fields; minimize sensitive content.
ActionDescription	TEXT	Plain-language event description.
ActionReason	TEXT	Reason for important changes, if required.
IPAddress	VARCHAR(45)	IP address where available.
UserAgent	VARCHAR(500)	Browser/device details, if available.
ActionStatus	VARCHAR(30)	Success or Failed.
Timestamp	DATETIME	Event timestamp.

Examples of events to log: login; student creation; profile modification; enrollment change; attendance edit; assessment score change; grade approval/posting; document verification; report generation/export; account or permission change; backup operation.

Audit integrity Audit records should be append-only for ordinary users. Avoid logging passwords, reset tokens, full payment/identity secrets, or unnecessary sensitive values. Restrict audit access and protect logs from alteration.

7.2 Notifications (Notifications)

Field	Suggested type	Purpose / notes
NotificationID	INT / UUID	Unique notification identifier.
UserID	INT / UUID	User receiving message, if the recipient has an account.
StudentID	INT / UUID	Related student, if applicable.
NotificationType	VARCHAR(50)	Attendance, Enrollment, Grade, Account, Profile, Document, etc.
NotificationTitle	VARCHAR(150)	Notification heading.
NotificationMessage	TEXT	Message body.
NotificationChannel	VARCHAR(30)	Portal, Email, SMS.
RelatedRecordID	VARCHAR(100)	Identifier for related attendance/enrollment/etc. record.
IsRead	BOOLEAN	Whether recipient has opened/read it.

Field	Suggested type	Purpose / notes
DateCreated	DATETIME	Creation timestamp.
DateRead	DATETIME	Read timestamp.
DeliveryStatus	VARCHAR(30)	Pending, Sent, Failed, Delivered.

Possible messages include: enrollment successfully verified; profile correction approved; document resubmission requested; absent for a scheduled class; new grades available. Notifications should not expose more personal or academic detail than necessary.

7.3 Backup configuration (BackupConfigurations)

Field	Suggested type	Purpose / notes
BackupConfigID	INT / UUID	Unique backup configuration.
BackupFrequency	VARCHAR(30)	Daily, Weekly, Monthly, or defined schedule.
BackupTime	TIME	Scheduled time.
BackupType	VARCHAR(30)	Full, Incremental, Differential.
StorageLocation	VARCHAR(500)	Protected backup destination/reference.
RetentionPeriodDays	INT	Number of days backups are retained.
IsEnabled	BOOLEAN	Whether schedule is active.
LastBackupDate	DATETIME	Last successful backup.
NextBackupDate	DATETIME	Next scheduled backup.

7.4 Backup history (BackupHistory)

Field	Suggested type	Purpose / notes
BackupID	INT / UUID	Unique backup operation identifier.
BackupConfigID	INT / UUID	Associated configuration.
BackupType	VARCHAR(30)	Type of backup performed.
BackupStartTime	DATETIME	Start timestamp.
BackupEndTime	DATETIME	Completion timestamp.
BackupSize	BIGINT	Backup size in bytes.
BackupFilePath	VARCHAR(500)	Protected backup storage reference.
BackupStatus	VARCHAR(30)	Pending, Running, Completed, Failed.
BackupChecksum	VARCHAR(255)	Integrity verification/checksum.
FailureReason	TEXT	Failure explanation, if applicable.
InitiatedBy	INT / UUID / SYSTEM	User or system process that initiated backup.

A backup plan should include secure storage, access control, recovery procedures, and regular restore testing. A successful backup job alone does not prove that the data can be restored.

7.5 Data retention policies (RetentionPolicies)

Field	Suggested type	Purpose / notes
RetentionPolicyID	INT / UUID	Unique policy identifier.
RecordType	VARCHAR(100)	Student Profile, Academic Record, Enrollment, Document, etc.
RetentionDuration	INT	Duration value.
RetentionUnit	VARCHAR(20)	Days, Months, Years, or Permanent.

Field	Suggested type	Purpose / notes
RetentionStartEvent	VARCHAR(50)	Graduation, Withdrawal, Record Creation, etc.
ArchiveAfterDays	INT	Number of days after the retention start event before archiving, if applicable.
ArchiveAt	DATE	Explicit archive date, if the policy uses a fixed calendar date.
DisposalAction	VARCHAR(30)	Archive, Secure Deletion, Permanent Retention.
IsActive	BOOLEAN	Policy status.
UpdatedBy	INT / UUID	Authorized updater.
UpdatedAt	DATETIME	Last modification.

Retention policy caution Do not hardcode a generic deletion period. Official academic records and student documents may have different institutional retention requirements. Confirm the applicable policy before implementing automated disposal.

7.6 System events and monitoring (SystemEvents)

Field	Suggested type	Purpose / notes
SystemEventID	INT / UUID	Unique system event identifier.
EventType	VARCHAR(50)	Application Error, Server Warning, Database Issue, Performance Alert, etc.
ModuleAffected	VARCHAR(100)	Module experiencing the event.
EventDescription	TEXT	Event details.
SeverityLevel	VARCHAR(20)	Information, Warning, Error, Critical.
EventTimestamp	DATETIME	Event timestamp.
ResolutionStatus	VARCHAR(30)	Open, Investigating, Resolved.
ResolvedBy	INT / UUID	ICT/EdTech user who resolved issue.
ResolutionDate	DATETIME	Date resolved.

Monitoring can include response time, error rate, database availability, job failures, and capacity indicators. It is primarily for ICT Services and Education Technology Services. These measurements help confirm that the system can handle expected peak enrollment and registration activity; collect them through monitoring infrastructure rather than manual user entry.

7.7 System performance metrics (SystemPerformanceMetrics)

Field	Suggested type	Purpose / notes
MetricID	INT / UUID	Unique performance measurement identifier.
MetricName	VARCHAR(100)	Name of the metric, e.g. response time, concurrent sessions, CPU utilization, database query duration.
MetricValue	DECIMAL(12,3)	Measured value.
MetricUnit	VARCHAR(30)	Milliseconds, count, percentage, MB, etc.
ComponentName	VARCHAR(100)	Application, database, API, reporting service, or server component measured.
MeasuredAt	DATETIME	Measurement timestamp.
ThresholdValue	DECIMAL(12,3)	Configured threshold, if applicable.
ThresholdStatus	VARCHAR(20)	Normal, Warning, Critical.
Environment	VARCHAR(30)	Production, Staging, Development, etc.

8. User Roles and Access Permissions

The scenario identifies ICT Services, Education Technology Services, Department Heads, Faculty Members, and the Registrar. The following matrix is a starting point for role-based access control (RBAC), not a substitute for final institutional authorization policy.

8.1 Permission legend

Label	Meaning
Full	Can manage the module within the limits of assigned administrative scope.
Create/Edit	Can create and modify permitted records.
View	Can view records within assigned scope.
Approve	Can review and approve/verify submitted records.
Restricted	Only specifically permitted functions or data are available.
None	No routine access.

8.2 Proposed role-permission matrix

Module / function	ICT Services	EdTech	Dept. Head	Faculty	Registrar
User authentication	Full	Restricted	View own	View own	Create/Edit account support
User account management	Full	Restricted	None	None	Restricted
Role and permission management	Full	Restricted	None	None	None
Student registration	Restricted	None	View	None	Full
Student profiles	Restricted / technical	View	View scoped	Restricted scoped	Full
Profile corrections	Restricted / technical	None	Review if delegated	None	Approve
Student documents	Restricted / technical	None	View if authorized	None	Full
Enrollment management	Restricted	Restricted	View/Edit scoped	View own classes	Full
Class assignment	Restricted	Edit support	Full	View assigned	View
Class scheduling	Restricted	Edit support	Full	View assigned	View
Attendance tracking	Restricted	None	View scoped	Create/Edit assigned	View
Grade management	Restricted / technical	None	Review/Approve	Create/Edit assigned	Approve/Finalize
Academic records	Restricted / technical	None	View scoped	Restricted scoped	Full
Reports and analytics	Full	Restricted	Departmental	Assigned classes	Full
Audit trails	Full	View	Restricted	None	Restricted
Backup management	Full	Restricted	None	None	None
System monitoring	Full	Full	None	None	None

Role separation ICT technical administrator access should not automatically mean routine authority to change official academic information. Sensitive administrator actions should be exceptional, justified, and audited. Faculty access should be limited to assigned classes; Department Head access should normally be scoped to their department.

8.3 Suggested permission codes

Permission code	Purpose
STUDENT_VIEW	View student information.

Permission code	Purpose
STUDENT_CREATE	Register a student record.
STUDENT_EDIT	Modify permitted student profile fields.
STUDENT_DEACTIVATE	Deactivate or archive a student record, subject to policy.
STUDENT_APPROVE	Approve student profile changes.
ENROLLMENT_VIEW	View student enrollment.
ENROLLMENT_CREATE	Create term enrollment.
ENROLLMENT_EDIT	Modify enrollment details.
ENROLLMENT_APPROVE	Verify enrollment.
ATTENDANCE_VIEW	View attendance.
ATTENDANCE_ENCODE	Record attendance.
ATTENDANCE_EDIT	Modify attendance entries.
GRADE_VIEW	View grades.
GRADE_ENCODE	Encode grades.
GRADE_EDIT	Modify submitted grades under controlled process.
GRADE_APPROVE	Approve grade submissions.
GRADE_POST	Finalize official grades.
REPORT_GENERATE	Generate reports.
REPORT_EXPORT	Export reports/documents.
USER_MANAGE	Manage accounts.
ROLE_MANAGE	Manage roles and permissions.
AUDIT_VIEW	View audit logs.
BACKUP_MANAGE	Configure and manage backup operations.
SYSTEM_CONFIGURE	Manage approved system configurations.

8.4 Scope and row-level access

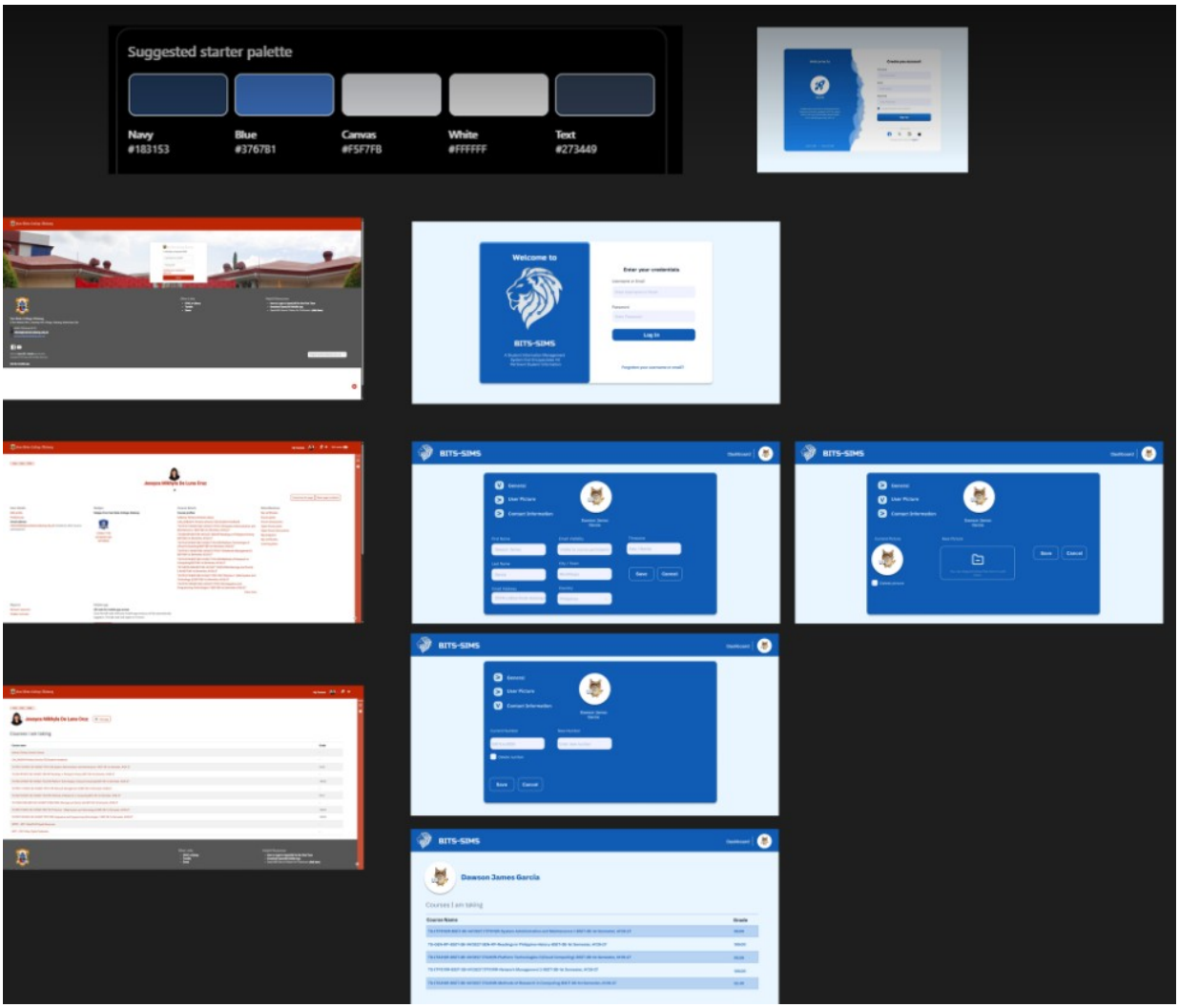
- A Faculty Member should only see the class lists, attendance, and grades for assigned ClassOfferings.
- A Department Head should be scoped to the department or departments assigned to them.
- A Registrar may have institution-wide access to student registration and official academic records as permitted by policy.
- Students should be limited to their own profile, enrollment, documents, grades, and attendance, plus approved profile-correction workflows.
- Technical administrator access should be separately managed and audited; avoid a blanket bypass for normal academic workflows.

8.5 Optional student self-service access

Function	Suggested Student permission
Own profile	View permitted profile details; submit correction request rather than directly changing official fields.
Enrollment and courses	View own current and prior enrollments and registered classes.
Grades	View own grades after the institution authorizes release.
Attendance	View own attendance records, where enabled.
Documents	View own submitted document status; upload or replace files only where the Registrar allows.
Notifications	View own portal notifications.
Other students / staff records	No access.

9. Recommended Website Pages and Interfaces

The blue mockups supplied for BITS-SIMS establish a blue-and-white visual direction with a login card, profile forms, upload controls, and a student course/grade listing. The red San Beda LMS screenshots are best treated as reference material for content density and familiar academic portal patterns, not as a source of BITS-SIMS branding.



Supplied UI reference collage: proposed color palette, blue BITS-SIMS login/profile mockups, and red LMS reference screens.

9.1 Visual design direction

Design token	Suggested value	Use
Navy	#183153	Brand/navigation/header background and strong emphasis.
Blue	#3767B1	Primary action buttons, active tabs, highlights.
Canvas	#F5F7FB	Main page background.
White	#FFFFFF	Cards, forms, content surfaces.
Text	#273449	Main text and labels.

9.2 Login page

Suggested fields and actions: Username/Email, Password, Remember Me, Forgot Password, Login, and an optional multi-factor challenge. Display the BITS-SIMS identity and brief system description on the branded panel.

9.3 Role-specific dashboard

Dashboard data

Field	Suggested type	Purpose / notes
TotalStudents	Calculated metric	Number of registered students.
ActiveStudents	Calculated metric	Number of students with Active status.
TotalDepartments	Calculated metric	Number of active departments.
TotalPrograms	Calculated metric	Number of active programs.
CurrentEnrollment	Calculated metric	Students enrolled in current academic term.
PendingEnrollment	Calculated metric	Enrollment records awaiting verification.
PendingProfileRequests	Calculated metric	Profile correction requests awaiting action.
PendingDocuments	Calculated metric	Documents awaiting verification.
AttendanceSummary	Calculated metric	Relevant attendance statistics for the user's scope.
CurrentAcademicTerm	Reference	Current academic year and semester.
RecentActivities	Audit-derived list	Recent permitted activities.

Dashboard content should vary by role. Faculty dashboards should prioritize assigned courses, today's schedules, attendance tasks, and grade submissions. Registrar dashboards should prioritize registration queues, enrollment, document verification, and student search. ICT/EdTech dashboards should emphasize accounts, monitoring, and system health.

9.4 Student management registry

Search/filter field	Purpose
StudentNumber	Search by official student number.
StudentName	Search by first name or surname.
DepartmentID / Department	Filter by department.
ProgramID / Program	Filter by academic program.
YearLevel	Filter by year level.
StudentStatus	Filter Active, Inactive, Graduated, Withdrawn, etc.
AdmissionType	Filter Freshman, Transferee, Returnee, etc.
AcademicYear	Filter by admission/enrollment year.

Suggested actions: Add Student, View Profile, Edit Information, View Enrollment, View Academic Records, Manage Documents, Review Profile Correction, and Deactivate/Archive Record.

9.5 Student profile page

Tab	Information shown
Overview	Student number, full name, program, department, year level, student status.
General Information	Personal and admission details.
Contact Information	Email, phone, and residential address.

Tab	Information shown
Guardian Information	Parent, guardian, and emergency contacts.
Enrollment History	Past and current term enrollments.
Courses	Subjects currently being taken.
Grades	Grades by subject and academic term.
Attendance	Attendance and tardiness history.
Documents	Submitted documents and verification status.
Academic History	Completed courses and prior academic performance.

9.6 Attendance interface

Field/control	Purpose
Academic Year / Semester	Select term.
Course / Section	Select faculty-assigned class.
Attendance Date	Select class meeting date.
Start Time	Show scheduled start time.
Student Number / Name	Identify student in roster.
Attendance Status	Present, Absent, Late, Excused, Left Early.
Time In / Time Out	Record arrival/departure, if used.
Remarks	Additional notes.
Save / Submit	Save or submit class attendance.

Show the full enrolled class roster in one table so faculty can record attendance for multiple students in one workflow.

9.7 Grade management interface

Field/control	Purpose
Academic Year / Semester	Select term.
Course / Section	Select assigned class.
Grading Period	Prelim, Midterm, Final, etc.
Assessment Name / Type	Identify quiz, examination, project, etc.
Student Number / Name	Identify student.
Assessment Score	Earned points.
Maximum Score	Maximum points, read from assessment definition.
Computed Percentage	Automatically calculated.
Final Grade	Computed/encoded final grade.
Remarks	Passed, Failed, Incomplete, etc.
Grade Status	Draft, Submitted, Approved, Posted.
Submit / Review History	Grade submission and approval workflow.

9.8 Reports page

Provide categories for Student Reports, Enrollment Reports, Academic Reports, Attendance Reports, and Departmental Reports. Support Generate, Preview, Download PDF, Export Excel/CSV, and Print, based on permissions and report type.

10. Data Validation and Business Rules

These rules should be enforced by the backend and database where possible. User interface validation improves usability but must not be the only control.

No.	Rule	Module
1	Every student must have a unique StudentNumber.	Registration
2	Usernames and account emails must follow uniqueness policy.	Authentication
3	A student must not have duplicate active term enrollments for the same AcademicTerm.	Enrollment
4	Students must be assigned to valid departments and programs.	Registration
5	Class enrollment must not exceed class capacity unless an authorized override is recorded.	Class Management
6	A room must not have overlapping active classes.	Scheduling
7	Faculty members should not have conflicting class assignments.	Scheduling
8	A student must not be enrolled twice in the same class offering.	Enrollment
9	Attendance must reference an enrolled student and a valid attendance session.	Attendance
10	Only assigned faculty or authorized staff may encode/modify attendance.	Attendance
11	Attendance must be unique per student per attendance session.	Attendance
12	Assessment scores must be within valid bounds, normally 0 through MaximumScore.	Grades
13	Assessment weights should total 100% when the grading scheme uses weighted components.	Grades
14	Grades must satisfy required review/approval before official posting.	Grades
15	Posted official grades require a controlled correction process.	Grades
16	Required documents must be verified according to the admission/enrollment rules.	Documents
17	Uploaded documents must pass file type, size, malware, and storage validation.	Documents
18	Only authorized reviewers may approve official profile corrections.	Profile Management
19	Changes to official student records should create an audit event.	Audit Trails
20	Archived academic records should remain retrievable to authorized personnel.	Retention

10.1 Computed values

Field / metric	Recommended calculation or source	Handling
StudentFullName	Compose FirstName + MiddleName + LastName + Suffix.	Display/computed; do not maintain as an independently editable copy.

Field / metric	Recommended calculation or source	Handling
Age	Current date minus DateOfBirth.	Compute when displayed; do not store as a value that becomes stale.
TotalRegisteredUnits	Sum Units for active ClassEnrollments in a term.	Compute or cache with clear invalidation.
TotalCompletedUnits	Sum units for course enrollments completed with passing outcomes under school rules.	Derive from official results.
AttendancePercentage	Applicable attended sessions divided by countable sessions.	Define treatment of excused absences and cancelled sessions.
FinalPercentage	Weighted calculation from scores and assessment weights.	Follow the institution's grading formula and rounding rule.
GPA / CumulativeGPA	Calculate from official grade points and the school's unit/credit rules.	Do not assume one universal formula.
CurrentClassSize	Count active ClassEnrollments for a ClassOffering.	Calculate rather than maintain manually where possible.
TotalActiveStudents	Count Students with current Active status.	Use a clearly defined status rule.

10.2 Suggested workflows

A. Applicant-to-student workflow

9. Applicant information is captured in Persons and AdmissionApplications.
10. Admission staff review required information and documents.
11. Accepted applicant is converted/linked to a Students record and assigned a StudentNumber.
12. Registrar verifies the student profile and creates or completes the first term enrollment.
13. Documents and enrollment state are updated; all important changes are logged.

B. Profile correction workflow

14. Student submits a correction request with field, existing value, requested value, reason, and evidence if required.
15. Registrar reviews the request and marks it Approved or Rejected with remarks.
16. Approved change is applied to the official profile and recorded in the audit trail.
17. Student receives a notification of the outcome.

C. Attendance workflow

18. Faculty selects an assigned ClassOffering and opens the applicable AttendanceSession.
19. System loads active class enrollments into the roster.
20. Faculty records Present, Absent, Late, Excused, or Left Early and optional times/remarks.
21. System validates duplicates and saves the attendance records.
22. Configured alert rules evaluate absence/tardiness and create notification jobs.

D. Grade workflow

23. Faculty creates assessments and defines maximum scores and weighting.
24. Scores are encoded for enrolled students and checked against maximum scores.
25. System calculates percentages and grading-period results according to configured rules.
26. Faculty submits grades; authorized reviewers review/approve as required.
27. Registrar or authorized role posts/finalizes official grades.
28. Corrections to posted grades follow a controlled, auditable process.

11. Database Inventory and Relationship Guide

The following inventory consolidates the main proposed tables. It is a reference map, not a requirement to expose every table as a separate user interface.

Area	Suggested table	Purpose / key fields
Identity	Persons	Shared person identity; PersonID, names, birthdate, contacts.
Student	Students	Student-specific record; StudentID, PersonID, StudentNumber, ProgramID, status.
Staff	Staff	Personnel details; StaffID, PersonID, EmployeeNumber, DepartmentID.
Authentication	Users	Login accounts; UserID, PersonID, Username, Email, PasswordHash.
Authorization	Roles	Role definitions; RoleID, RoleName, RoleCode.
Authorization	Permissions	Action permissions; PermissionID, PermissionCode, ModuleName.
Authorization	UserRoles	Users-to-roles assignment.
Authorization	RolePermissions	Roles-to-permissions assignment.
Organization	Departments	Department structure.
Organization	Programs	Academic programs linked to departments.
Academic setup	AcademicTerms	Academic years and semesters.
Academic setup	Courses	Subject catalogue.
Academic setup	CoursePrerequisites	Course prerequisite relationships.
Class setup	ClassOfferings	Course section/term offerings.
Class setup	ClassSchedules	Meeting days, times, and room.
Class setup	Rooms	Room catalogue.
Class setup	FacultyAssignments	Faculty-to-class assignment.
Enrollment	StudentEnrollments	Overall student enrollment by term.
Enrollment	ClassEnrollments	Student-to-class enrollment.
Admission	AdmissionApplications	Applicant and admission workflow.
Profile	Guardians	Parent/guardian contacts.
Profile	EmergencyContacts	Emergency contact details.
Profile	EducationalBackground	Prior education details.
Profile	ProfileCorrectionRequests	Student profile change requests.
Documents	DocumentTypes	Configurable required document categories.
Documents	StudentDocuments	Uploaded document metadata and status.
Documents	DocumentVerifications	Review history.
Attendance	AttendanceSessions	One class meeting.
Attendance	AttendanceRecords	One student attendance result per session.
Attendance	AttendanceAlerts	Absence/tardiness alerts.
Attendance	AttendanceAlertSettings	Configurable alert rules.
Grades	GradingPeriods	Prelim/Midterm/Final or configured periods.
Grades	Assessments	Quizzes, examinations, projects, and weights.
Grades	AssessmentScores	Student scores per assessment.
Grades	StudentGrades	Period/final official grades.
Grades	GradeScales	Configured grade equivalences.
Academic history	StudentAcademicRecords	Term-level academic summary/cache.
Reporting	ReportDefinitions	Report templates and permission metadata.
Reporting	GeneratedReports	Report generation jobs and file references.
Reporting	ReportHistory	Report generation/download history.

Area	Suggested table	Purpose / key fields
Security	AuditLogs	Append-only user and system activity records.
Notifications	Notifications	Portal/email/SMS notification records.
Operations	BackupConfigurations	Backup schedule and retention settings.
Operations	BackupHistory	Backup execution history.
Operations	RetentionPolicies	Record retention and disposal rules.
Operations	SystemEvents	Errors, warnings, and operational events.
Operations	SystemPerformanceMetrics	Response time, concurrent sessions, utilization, and component performance measurements.

11.1 Core relationships

Parent / source	Relationship	Child / target
Persons	1 : 0..1 / 1 : many depending on identity model	Students and Staff; a person can have a student and/or staff association where policy permits.
Departments	1 : many	Programs, Students, Staff, ClassOfferings.
Programs	1 : many	Students and ClassOfferings.
AcademicTerms	1 : many	StudentEnrollments, ClassOfferings, StudentGrades, GradingPeriods.
Courses	1 : many	ClassOfferings; also self-references through CoursePrerequisites.
ClassOfferings	1 : many	ClassSchedules, FacultyAssignments, ClassEnrollments, AttendanceSessions, Assessments.
Students	1 : many	StudentEnrollments, ClassEnrollments, AttendanceRecords, StudentDocuments, Guardians, Notifications, StudentGrades.
StudentEnrollments	1 : many	ClassEnrollments.
AttendanceSessions	1 : many	AttendanceRecords.
Assessments	1 : many	AssessmentScores.
ClassEnrollments	1 : 0..many	StudentGrades; depending on whether period and final grades are stored as separate records.
Users	many : many	Roles through UserRoles.
Roles	many : many	Permissions through RolePermissions.

11.2 Simplified relationship view

PERSONS - shared identity record
**STUDENTS	STAFF	USERS**
STUDENT ENROLLMENTS -> CLASS ENROLLMENTS
CLASS OFFERINGS -> COURSES / SCHEDULES / FACULTY ASSIGNMENTS
ATTENDANCE SESSIONS -> ATTENDANCE RECORDS
ASSESSMENTS -> ASSESSMENT SCORES -> STUDENT GRADES
**DOCUMENTS	GUARDIANS	PROFILE CORRECTIONS	REPORTS**
**AUDIT LOGS	NOTIFICATIONS	BACKUPS	RETENTION	MONITORING**

Relationship note: Departments connect to Programs; Programs and AcademicTerms influence student and class assignment; Courses are offered through ClassOfferings. All student-facing academic transactions resolve back to one StudentID.

11.3 Important normalization decisions

- Keep Persons separate from Students and Staff so shared personal identity is not duplicated.
- Keep Users separate from Students and Staff. A user account grants login access; it is not the institutional student/staff record itself.
- Use junction tables for many-to-many assignments, especially UserRoles, RolePermissions, ClassEnrollments, and FacultyAssignments.
- Separate Courses (catalogue subject) from ClassOfferings (a specific section in a specific term).
- Separate overall StudentEnrollment from ClassEnrollment (the selected subjects/sections).
- Separate AttendanceSessions (class meeting) from AttendanceRecords (student result for that meeting).
- Separate Assessments from AssessmentScores and official StudentGrades.
- Use StudentDocuments metadata to reference files in secure storage; keep verification history separately if multiple reviews are needed.
- Keep official grades immutable after posting except through a controlled correction process that records old/new values and approver.

12. Recommended Development Priorities

The system is extensive. A phased approach reduces integration risk and ensures that the central student and enrollment records are stable before advanced analytics and operations are added.

Phase 1 - Core system

- Login and authentication
- User account management
- Role-based access control
- Department and program management
- Student registration and profile management
- Student enrollment and class enrollment
- Course and class offering management
- Basic academic record management

Phase 2 - Academic functions

- Class schedule and faculty assignment management
- Attendance sessions and attendance tracking
- Attendance alert rules and notifications
- Assessment and score management
- Grade computation, review, approval, and posting
- Student document management and verification
- Profile correction requests
- Student academic history and profile tabs

Phase 3 - Reporting, security, and reliability

- Reports and analytics
- Printable student report cards and transcripts
- Departmental and attendance reports
- Audit trail viewing and controls
- Notification delivery monitoring
- Automated backup and recovery operations
- Data retention and archival management
- System monitoring and performance review

12.1 Suggested minimum viable release

For an initial working demonstration, prioritize a complete end-to-end path: Registrar creates a student -> student is enrolled in an academic term -> student is assigned to one or more class offerings -> Faculty sees their assigned class list -> Faculty records attendance and grades -> Registrar can view the student academic summary and generate a basic report. This validates the most important relationships before expanding the system.

12.2 Suggested development deliverables

Deliverable	What it should contain
Requirements traceability	Mapping from each scenario requirement to one or more modules, pages, tables, and test cases.
ERD	Entities, primary keys, foreign keys, cardinalities, unique constraints.
Data dictionary	Final field names, data types, nullability, defaults, constraints, and definitions.
Permission matrix	Role-by-role permissions, department scope, student self-service scope.
Workflow diagrams	Admission, enrollment, document review, profile correction, attendance, and grade posting.
UI wireframes	Role-specific dashboards, forms, table views, detail pages, and report flows.
Validation test cases	Unique identifiers, duplicate enrollment, schedule conflict, grade bounds, authorization, and audit behavior.
Deployment and recovery plan	Environment configuration, backup schedule, restore test, access control, and monitoring.

Appendix A - Coverage of the Original Scenario

This table maps the key requirements in the supplied Scenario 9 brief to the proposed system design in this reference.

Scenario requirement	Proposed components / coverage
Admissions and Enrollment - new student applications, registration, class assignments	AdmissionApplications; Persons; Students; StudentEnrollments; ClassEnrollments; ClassOfferings.
Attendance Tracking - daily attendance, tardiness, absence alerts	AttendanceSessions; AttendanceRecords; AttendanceAlerts; AttendanceAlertSettings; Notifications.
Grade Management - encode grades, final scores, report cards	GradingPeriods; Assessments; AssessmentScores; StudentGrades; GradeScales; reports.
Reporting Tools - class lists, demographic summaries, performance charts	ReportDefinitions; GeneratedReports; ReportHistory; student, academic, attendance, and departmental reports.
Student Profiles - personal details, contact information, academic history	Persons; Students; Guardians; EmergencyContacts; EducationalBackground; StudentAcademicRecords.
Document Management - digital storage of entry requirements	DocumentTypes; StudentDocuments; DocumentVerifications.
Audit Trails - track edits to student information	AuditLogs for create/read/update/delete, approvals, exports, and relevant system events.
Data Backup - automatic backups	BackupConfigurations; BackupHistory; tested restore procedures.
Security and Privacy - role-based permissions and sensitive data protection	Users; Roles; Permissions; UserRoles; RolePermissions; scoped server-side authorization.
System Performance - quick operation during peak periods	SystemEvents/monitoring; indexed queries; pagination; efficient report generation and database tuning.
Data Retention - retain records based on institutional requirements	RetentionPolicies; archive/disposal workflows; preservation of official academic history.

Appendix B - Assumptions and decisions to confirm

- The institution has a defined official StudentNumber format and rules for generating or assigning it.
- The school will confirm the exact admission types, student statuses, year-level conventions, and program structure.
- The academic calendar and grading scheme (period names, weights, rounding rules, passing thresholds, GPA calculations) must be supplied by the institution.
- The school will decide whether students have direct portal access to view grades, attendance, documents, and profile details.
- The school will decide whether absence alerts go to faculty, parents/guardians, students, or a subset, and what thresholds trigger them.
- The school will define which personnel can approve enrollment, official profile corrections, grade posting, transcripts, and document verification.
- The school will determine document formats, maximum file sizes, retention periods, and approved storage location.
- The school will decide whether BITS-SIMS integrates with institutional email, SMS gateway, existing LMS, or other student systems.
- The system will follow the school's approved privacy, records-management, and information-security policies.

Appendix C - Implementation notes and cautions

- Do not interpret every suggested field as mandatory. Collect only fields that have a defined operational or institutional purpose.
- Use lookup tables for values that may change, such as student status, document type, admission type, course type, notification channel, and grade remarks.
- Use database constraints for uniqueness and relationships, and validate business rules in backend services as well.

- Treat the access matrix as an initial proposal. Test access with one account per role and verify both allowed and denied actions.
- Keep user-facing pages simple and role-specific even though the underlying data model is extensive.
- Avoid creating duplicate copies of full student data in grade, attendance, report, or class-list tables. Reference the central records instead.
- Where snapshots are needed for historical reports, label them as snapshots and define when/how they are captured.
- For sensitive records, apply access controls, audit access, use secure storage, and avoid excessive data in exported files.

Appendix D - Glossary

Term	Meaning in this reference
Primary key (PK)	Field that uniquely identifies a row in a table.
Foreign key (FK)	Field that references a primary/unique key in another table.
Junction table	Table used to represent a many-to-many relationship.
RBAC	Role-Based Access Control: permissions are assigned to roles and roles to users.
ClassOffering	Specific instance of a course offered in a section during an academic term.
StudentEnrollment	Student's overall registration for one academic term.
ClassEnrollment	Student's registration in one specific ClassOffering.
AttendanceSession	One meeting/class date for which attendance is recorded.
Assessment	Graded activity such as quiz, exam, project, or laboratory task.
Official grade	Grade that has completed the required review and posting process.
Audit trail	Protected history of actions performed in the system.
Data retention	Rules determining how long records are kept, archived, or securely disposed of.

Closing design principle BITS-SIMS should have one authoritative student identity record, linked enrollment and class records, controlled academic transactions, and report outputs generated from those trusted records. The field catalogue should be refined with institutional policies before database implementation.